

練習プリント 02 解答

式・単位・中間計算を確認する

1. $\omega = 2\pi N/60 = 2\pi \times 1500/60 = 157 \text{ rad/s}$
2. $\omega = 2\pi \times 1800/60 = 188.5 \text{ rad/s}$, $P = \omega T = 188.5 \times 2.0 = 377 \text{ W}$
3. $\omega = 2\pi \times 1500/60 = 157.1 \text{ rad/s}$, $T = P/\omega = 3000/157.1 = 19.1 \text{ N}\cdot\text{m}$
4. $N_{\text{out}} = N_{\text{in}}/i = 3000/15 = 200 \text{ min}^{-1}$
5. $T_{\text{out}} = T_{\text{in}} \times i = 0.80 \times 20 = 16 \text{ N}\cdot\text{m}$
6. $T_{\text{out}} = 0.80 \times 20 \times 0.80 = 12.8 \text{ N}\cdot\text{m}$
7. $v = \pi DN/60 = \pi \times 0.20 \times 300/60 = 3.14 \text{ m/s}$
8. $F = T/r = 5.0/0.10 = 50 \text{ N}$
9. $\eta = P_{\text{out}}/P_{\text{in}} \rightarrow P_{\text{in}} = 1.6/0.80 = 2.0 \text{ kW}$
10. $P_{\text{loss}} = 2.5 - 2.0 = 0.50 \text{ kW} = 500 \text{ W}$, $\eta = 2.0/2.5 \times 100 = 80\%$
11. $\eta_{\text{total}} = 0.90 \times 0.80 = 0.72 = 72\%$
12. $E = Pt = 500 \times (3.0 \times 60) = 90,000 \text{ J} = 90 \text{ kJ}$